PRAKHAR SINGH

Machine Learning / AI Engineer

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PROFILE

ML/AI engineer with strong foundations in machine learning, deep learning, probability, statistics, and linear algebra. Built projects spanning predictive maintenance, autonomous AI systems, and end-to-end AI applications.

PROJECTS

GoalOps - Autonomous Business Goal Operator | LLMs | MCP | PostgreSQL
github.com/iamprakhar10/goalops

- Built an AI powered operator that observes a simulated SaaS business, selects interventions, takes action and evaluates progress toward a goal.
- Exposed predefined business capabilities through MCP tools.
- Designed reproducible simulation runs for better evaluation.

Public Pulse - AI Civic Assistance Platform | FastAPI | PostgreSQL | LangGraph | OAuth 2.0 | JWT
github.com/iamprakhar10/Public-Pulse

- Built a civic complaint platform that brings people and government authorities closer and makes them more responsible. It also helps journalists to report better and be effective.
- Implemented LangGraph workflow, Google OAuth2.0, JWT authentication, SQLAlchemy persistence, REST API and streamlit for frontend.
- Used pincode, location, authority, city and state for complaint. Arbitrary locations are not supported.

Turbofan Remaining Useful Life Prediction
github.com/iamprakhar10/aircraft-rul-predictor

- Built end-to-end RUL prediction pipeline on NASA C-MAPSS turbofan sensor data, comparing gradient boosted models with LSTMs.
- Built capped-RUL targets, sliding-window sequences, and condition-aware normalization for multi-condition FD004 data. Achieved 9.64 MAE on FD001 test data.
- Established that LSTM is actually benefiting from temporal sequences rather than just reducing the error.

TECHNICAL SKILLS

ML / Deep Learning: PyTorch, scikit-learn, regression/classification, boosting, PCA, GMM/EM, RNN/LSTM/GRU, Transformer,Attention

GenAI / Engineering: LLMs, RAG, Agentic AI, LangChain, LangGraph, MCP, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Pydantic, JWT, OAuth 2.0, Git, Python, AWS

Math: Statistics, Statistical Inference, Probability, Linear Algebra, Optimization

EDUCATION & COURSEWORK

Indian Institute of Technology, Kanpur | UG Studies, 2018-2020

JEE Advanced 2018 AIR 4515 | **JEE Mains** AIR 4042 | **International Math Olympiad** (2014-15) AIR 97

Independent Study: Stanford CS229 | Cornell CS4780 | CMU 11-785 | MIT 18.06 | MIT 18.650 | Harvard Stat 110